

PancakeSwap: Supporting the DeFi ecosystem with an always- available infrastructure

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PancakeSwap is on Google Cloud to make the platform available 24/7 so that users can trade seamlessly at any time.

About PancakeSwap

PancakeSwap is a DeFi trading platform that aims to simplify trading on the blockchain, making it a seamless experience for all. It aims to remain at the forefront of DeFi innovation by always providing users a top-notch experience.

Industries: Financial Services

Location: Singapore

(<https://cloud.google.com/kubernetes-engine/>),

Google Cloud results

- Helps users get connected to BNB endpoints at any time with load balancing
- Enables full platform scalability to support high traffic
- Accurately predict spikes of traffic surge, resulting in more than 30% of cost savings and slashing the peak response

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About Helix Technologies

With a mission to help brands accelerate innovation, Helix Technologies provides organizations with cloud migration strategies and execution expertise.



Google Cloud (<https://cloud.google.com/>)

Google Cloud Armor
(<https://cloud.google.com/armor/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Operations
(<https://cloud.google.com/products/operations>)

Logging (<https://cloud.google.com/logging/>)

Monitoring (<https://cloud.google.com/monitoring/>)

time by

62.5x

- Works with audit firms to ensure there are no loopholes
- Utilizes Cloud Armor to filter sensitive data

The finance world has been transformed with the emergence of decentralized finance (DeFi) in recent years. While blockchain technology has been around for a while, the cryptocurrency world grew quickly with the introduction of non-fungible tokens (NFTs), bringing the concept of blockchain trading closer to the masses.

PancakeSwap (<https://pancakeswap.finance/>) is among the many trading platforms available to most users around the world, where people can trade their assets without any intermediary. "The platform allows people to trade anytime they want, or exchange their cryptocurrency," says Chef Jojo, Technical Lead at PancakeSwap. In fact, the name PancakeSwap was born as a way to make what would otherwise be seen as a difficult to understand financial service, more relatable.

According to Chef Jojo, whose real name is not used to foster a permissionless ecosystem, the playful and lighthearted vibe is also reflected in the various product themes. "The decision to keep our contributors anonymous is a strategic move to promote equal participation and minimize personal bias," says Chef Jojo. "By focusing on the project rather than personal recognition or reputation, the team can prioritize the needs and interests of the PancakeSwap community as a whole."

The team is focused on providing a user-friendly DeFi application where people can trade even without much technical or blockchain knowledge. "We don't hold any funds for users as it's all in their own wallets," says Chef Jojo. The platform also offers gamification features, where people can play around with DeFi.



Turning to Google Cloud for a more reliable infrastructure

Being such a volatile market, it is essential that PancakeSwap maintains high availability at all times. "When we initially launched, we offered just a simple application on the frontend that connected directly to the BNB Chain. However, as the number of users started to grow, and we saw higher traffic, the simple infrastructure wasn't enough anymore," Chef Jojo explains.

For that reason, it worked with Helix Technologies (<https://www.helixtechnologies.xyz/>), a Google Cloud (<https://cloud.google.com/>) partner who helped build its infrastructure and implement the appropriate cloud solutions to support PancakeSwap's evolving needs. As a priority, PancakeSwap needed to ensure that users could always get connected to the platform. With that in mind, the team used Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to properly route traffic to regional kubernetes clusters, allocating users to different endpoints, to better spread out traffic.

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Supporting a fluctuating number of requests with high scalability

As a trading platform that gives full autonomy to
users, being always available is crucial for
PancakeSwap. For that reason, the team leverages

Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) to scale its
nodes quickly when necessary.

"With Google Kubernetes Engine, we can quickly scale up within seconds when there are many requests," The team also leverages Operations Suite (<https://cloud.google.com/products/operations>), particularly Cloud Logging (<https://cloud.google.com/logging>) and Cloud Monitoring (<https://cloud.google.com/monitoring>) to ensure platform stability at all times. With that, the BNB chain is able to manage even a large amount of requests without any downtime, maintaining an uptime of 99.99%.

Along with Google Kubernetes Engine, PreditKube solution from Dysnix accurately predicted more than 90% of traffic spikes on PancakeSwap, and automated up-and-down scale of blockchain nodes ahead of time to properly manage the anticipated traffic surge. This resulted in more than 30% of cost saving and slashed the peak response time by 62.5 times.

Since implementing Google Cloud solutions, the average request time to a node has been reduced to 100ms, resulting in faster and more reliable service for users. Using the fully managed services of Google Cloud also means that less time is spent dealing with infrastructure issues, so engineers can devote their time on developing new features and improving the user experience.

"At one point, we had more than one billion requests on the BNB Chain, but this is changing all the time due to the BNB network changes, among other variables. With Google Kubernetes Engine, we can quickly scale up when there are many requests."

—Chef Jojo, Technical Lead, PancakeSwap

Maintaining a secure trading environment to keep users coming back

As with any trading platform, the most vulnerable space where major hacks happen is in the smart contract, where a code is written into the blockchain to execute tasks that would typically be handled by a person. For cryptocurrency trading to be truly decentralized, smart contracts are necessary. For that reason, the PancakeSwap team spends a lot of time ensuring that their smart contracts are as secure as possible.

"We work with a lot of audit firms to make sure there are no loopholes, and use Cloud Armor (<https://cloud.google.com/armor?hl=en>) to filter sensitive data," says Chef Jojo.

Having launched a few successful products on its platform, the next steps for PancakeSwap is to start looking into improving the trading experience. "We are looking to expand our services to include more chains, and explore non-fungible tokens (NFT) and other things to give our users more options," says Chef Jojo.

The team is looking at the possibility of leveraging BigQuery (<https://cloud.google.com/bigquery>) to manage and analyze the blockchain historical data as the company grows. "More and more, we are noticing that users want to receive meaningful notifications on the blockchain. Even though everything on the blockchain is public, a lot of times the data and information can be quite technical with raw transactions and a lot of metadata," explains Chef Jojo. With BigQuery, PancakeSwap aims to simplify this data, and present it in a way that can be easily understood by the layman.

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